Study methodology

Sources: mission specification · URDF chain
Wheel SET: branch · arm mapper (5 aliases / 4 channels)

A Coordinator
28,672 pairs / policy
Oracle + BFS; 109 checks

→ **0 full-policy violations**
Ablated guard violations:
480 / 128 / 288

B Wrist geometry
68,921 joint poses
1,013 independent FK poses

→ **0.200 m link-sum bound**
2,747 below ground
$\Delta \leq 6.8 \times 10^{-17}$ m

C Command interfaces
261,121 wheel pairs
2,255 arm-mapping cases

→ **1,020: no command**
Mixed-zero wheel pairs
Magnitude lost; 29 checks

Needs: stop feedback · calibrated reach · wheel-rate semantics